Q8. Self-Organizing Map (SOM)

Manual Solution

Given

Parameter	Value
Number of clusters (neurons)	3
Number of epochs	50
Learning rate	0.5
Data centres	(1,1), (5,5), (1,5), (5,1)

Step 1: Initial Weights

The initial weights are picked at random from the generated data points.

Neuron	Initial Weight
N0	(0.269, 1.695)
N1	(5.660, 5.341)
N2	(0.749, 4.878)

Step 2: Distance Calculation

Take the first point **P = (0.269, 1.695)**. The squared distance to a neuron with weight W = (w_x, w_y) is:

$$d(P, W) = (x - w_x)^2 + (y - w_y)^2$$

Neuron	Calculation	Distance
N0	(0.269 - 0.269)^2 + (1.695 - 1.695)^2	0.000
N1	(0.269 - 5.660)^2 + (1.695 - 5.341)^2	42.356
N2	(0.269 - 0.749)^2 + (1.695 - 4.878)^2	10.362

The smallest distance is 0, so **N0 is the winning neuron**.

Step 3: Weight Update

Update rule:

$$W_{new} = W_{old} + (\text{Learning Rate}) \times (P - W_{old})$$

Winner N0: the point and the weight are identical, so (P – W) = 0 and the weight stays at (0.269, 1.695).

Neighbour N1: its learning rate is halved: 0.5 / 2 = 0.25.

$w_x = 5.660 + 0.25 \times (0.269 - 5.660) =$ **4.312**

$w_y = 5.341 + 0.25 \times (1.695 - 5.341) =$ **4.430** (4.4295 rounded)

Updated weight of N1: **(4.312, 4.430)**

The same process is repeated for every generated point over 50 epochs.

Step 4: Final Output

Cluster	Points	Final Weight
Cluster 0	15	[0.91, 0.86]
Cluster 1	15	[1.04, 5.08]
Cluster 2	15	[5.13, 4.97]

Result

The SOM forms three clusters of 15 points each. The final neuron weights settle near the data centres (1,1), (1,5) and (5,5), showing that updating weights by distance moves each neuron toward the group of points it represents.